Assignment 3

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Notion link: https://www.notion.so/Assignment-3-1e6ee40c6a08802a88dbe4ac0ae67207

1. Write an if statement that assigns 1 to x if y is greater than 0.

```
import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        int x;
        Scanner input = new Scanner(System.in);
        System.out.print("Assign a number to y: ");
        int y = input.nextInt();
        if (y > 0) {
            x = 1;
        }
    }
}
```

2. Write an if statement that increases pay by 3% if score is greater than 90, otherwise increases pay by 1%.

```
import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        double pay;
        Scanner input = new Scanner(System.in);
        System.out.print("Enter your score: ");
        int score = input.nextInt();
        if (score > 90) {
            pay*= 1.03;
        } else {
            pay*= 1.01;
        }
    }
}
```

3. What is the output of the code in (a) and (b) if number is 30? What if number is 35?

    a.

```
30 is even
30 is odd

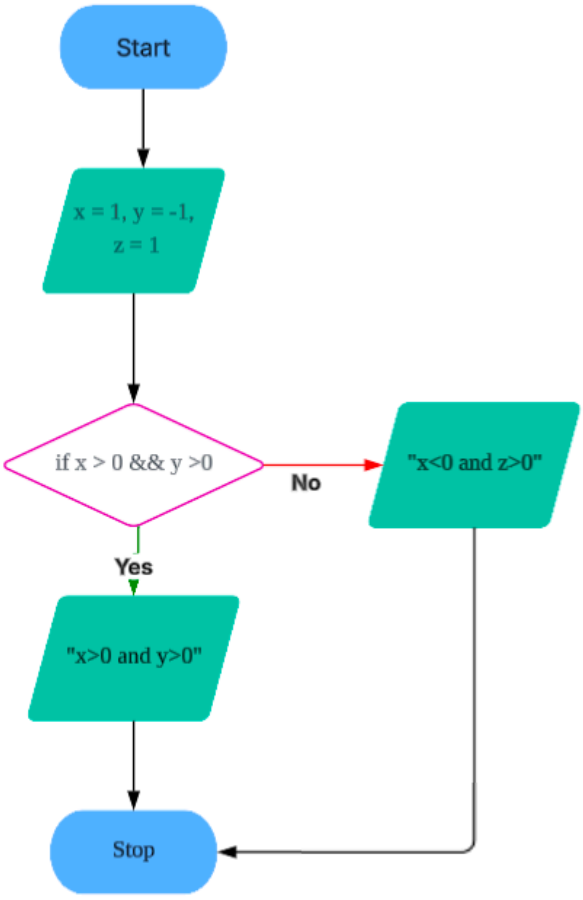
35 is odd
```

b.

```
30 is even
35 is odd
```

4. Suppose x = 1, y = -1, and z = 1. What is the output of the following statement? Draw a flowchart of the code.

```
x < 0 and z > 0
```



5. Assuming the input is a valid member no, trace the following algorithm if memberNo = “Member3” and amount = 999.

No.	Code	memberNo	Amount	disount	total	output
1	Start					
2	Read memberNo	Member3				
3	Read amount		999			
4	If (memberNo is valid) then					true
5	If (amount < 100) then					false

6	discount = 0.02					
7	Else if amount(<500) then					false
8	discount = 0.05					
9	Else discount = 0.10			0.10		true
10	Total = amount - (amount*discount)				899.1	
11	Print memberNo					Member3
12	Print total					899.1
13	End					

6. Convert the following if-else-if statement into a switch statement.

```
switch (choice) {
   case 1:
      System.out.println("You selected 1.");
      break;
   case 2:
   case 3:
      System.out.println("You selected 2 or 3");
      break;
   case 4:
      System.out.println("You selected 4.");
      break;
   default:
      System.out.println("Select again please.");
   }
```

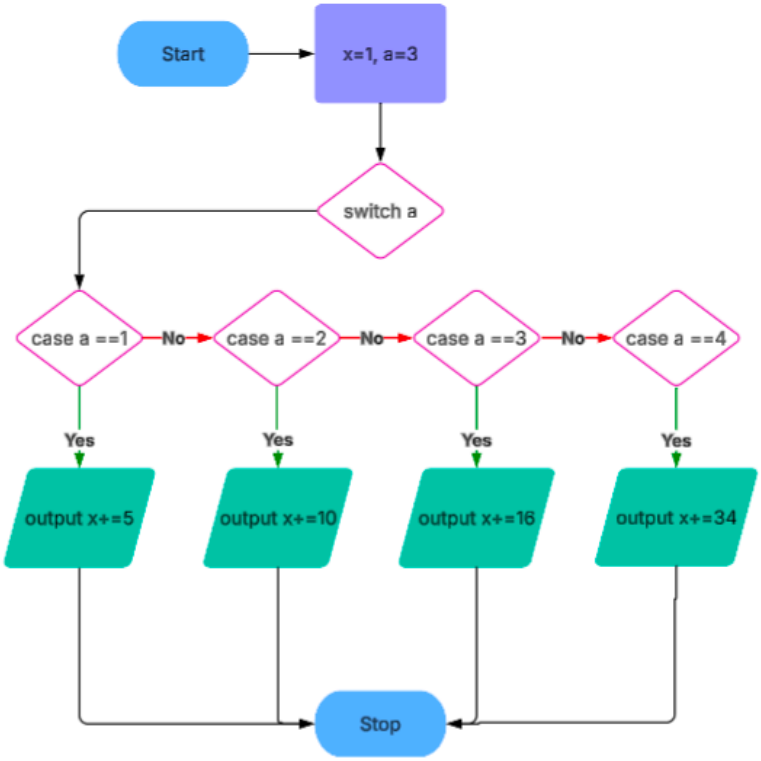
7. Write the conditional expression based on the if-else statement below:
   i. q = (x < y) ? 0 : 1;
   ii. q = (x < y) ? (a + b) : (x + 2);
   iii. System.out.println(number + ((number % 2 == 0) ? " is even" : " is odd"));
8. Analyze the following program fragment and give your answer.

   x = 2
9. What is x after the following if-else statement is executed? Use a switch statement to rewrite it and draw the flowchart for the new switch statement.

   x = 17

   ```
   int x = 1, a = 3;
   switch (a) {
      case 1:
         x += 5;
         break;
      case 2:
   ```

```
        x += 10;
        break;
    case 3:
        x += 16;
        break;
    case 4:
        x += 34;
```



10. Produce problem analysis chart (PAC), input-processing-output (IPO) chart, flowchart, and Java program for the following problem.

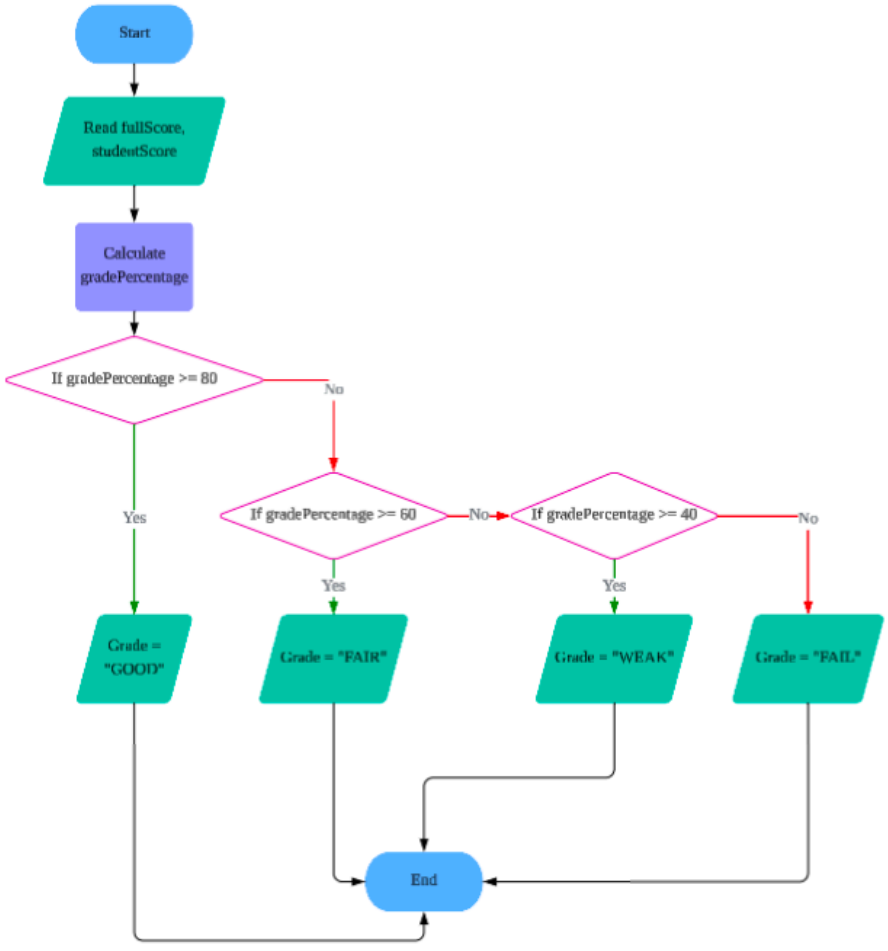
PAC

Data	Processing	Output
fullScore, studentScore	Calculate gradePercentage if (gradePercentage ≥ 80) status = “GOOD” else if (gradePercentage ≥ 60) status = “FAIR” else if (gradePercentage ≥ 40) status = “WEAK” else status = “FAIL”	gradePercentage, status

IPO

Input	Process	Output
fullScore, studentScore	- Read fullScore, studentScore - Calculate gradePercentage = studentScore/fullScore - If (gradePercentage ≥ 80) status = “GOOD” Else if (gradePercentage ≥ 60) status = “FAIR” Else if (gradePercentage ≥ 40) status = “WEAK” Else status = “FAIL” Endif	gradePercentage, status

Flowchart



Java code

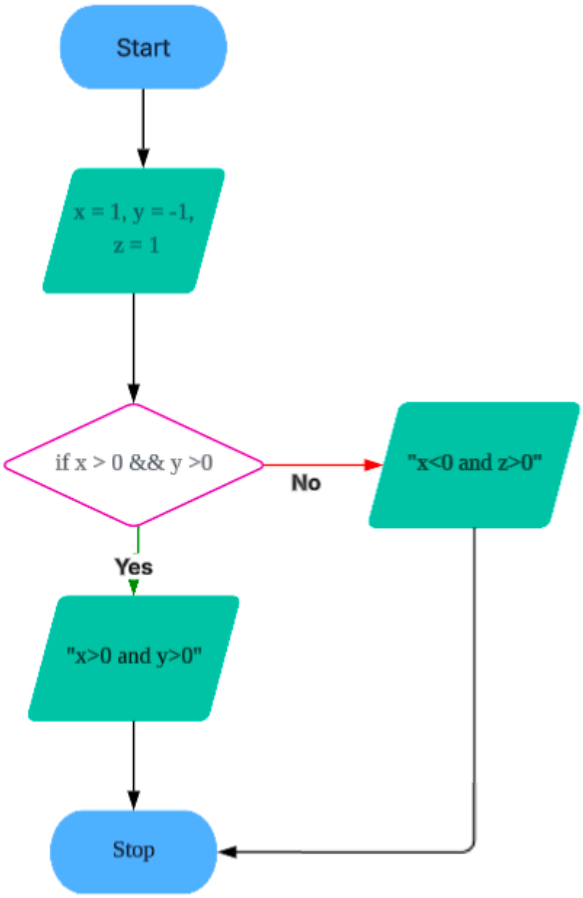
```
import java.util.Scanner;

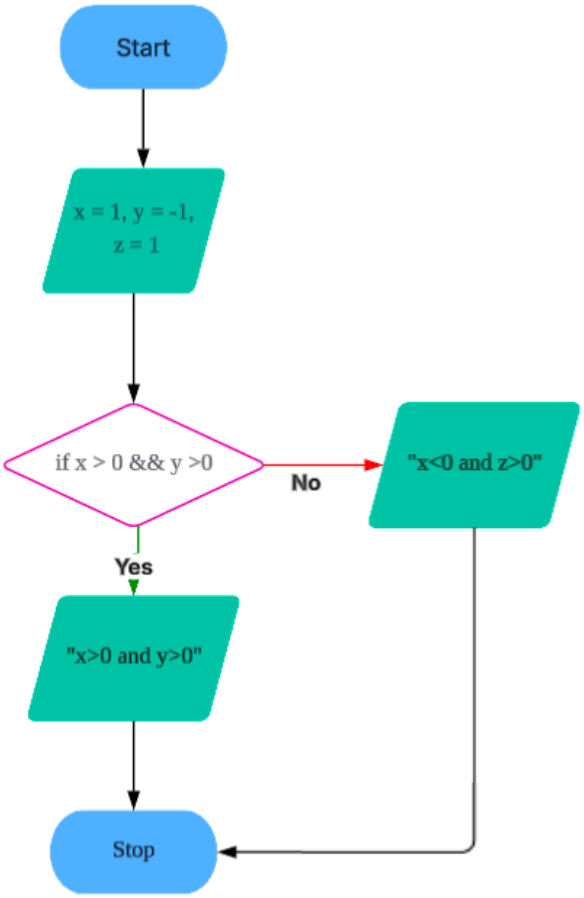
class Main {
    public static void main(String[] args) {
```

```
        Scanner input = new Scanner(System.in);
        System.out.println("Enter full score: ");
        int fullScore = input.nextInt();
        System.out.println("Enter student's score: ");
        int studentScore = input.nextInt();
        double gradePercentage = studentScore * 100.0 / fullScore;

        String status;

        if (gradePercentage >= 80) {
            status = "GOOD";
        } else if (gradePercentage >= 60) {
            status = "FAIR";
        } else if (gradePercentage >= 40) {
            status = "WEAK";
        } else {
            status = "FAIL";
        }
        System.out.println("Student's grade percentage is " + gradePercentage + "(" + status + ")");
    }
}
```


Start
x = 1, y = -1,
z = 1
if x > 0 && y >0
No
"x<0 and z>0"
Yes
"x>0 and y>0"
Stop


Start
x = 1, y = -1,
z = 1
if x > 0 && y >0
No
"x<0 and z>0"
Yes
"x>0 and y>0"
Stop